

REFLECTIVE OPTICAL PHOTOPLETHYSMOGRAPHY (cPPG)

Mathematical Formulas, Signal Processing Pipeline & Peer-Reviewed Scientific Validation

Project: SMRUTI Vitality Health Checkup Suite | **Sensing Modality:** Camera-based Reflective PPG (cPPG) | **Release:** Technical & Scientific Analysis Report
Scope: Mathematical inventory of all algorithmic formulas, source files, physiological foundations, and direct literature citations.

1. System Architecture & The Optical Extraction Process

Hardware Isolation & The Sensor Transition: Modern smartphone fingerprint sensors operate exclusively within a hardware-isolated **Trusted Execution Environment (TEE)** (e.g. ARM TrustZone on Android via IBiometricsFingerprint.hal or Apple Secure Enclave). Third-party user-space applications are cryptographically prohibited from accessing raw capacitance matrices or optical sensor frames. To deliver non-invasive health screening on commodity smartphones, this system transitions to **Reflective Optical Photoplethysmography (cPPG)** using the rear CMOS camera sensor combined with the adjacent continuous LED flash.

Stage	Module / File	Engineering Operation	Physiological / Signal Objective
1. Optical Capture	PulseSheet.tsx CameraView	Captures camera frames at 10?15 FPS with flash enabled, skipProcessing=true, and repair of truncated JPEG EOI markers (0xFF, 0xD9).	Transilluminates capillary beds of fingertip dermis; captures time-series optical backscatter without OS camera orientation delays.
2. Tissue Verification	pulse.ts frameBrightness()	Subsamples frame red and green channels. Enforces avgRed ? 95 and avgRed ? 1.15 × avgGreen.	Prevents mathematical aliasing from 50/60 Hz ambient AC lighting that otherwise aliases into false 52 BPM signals in open air.
3. Temporal Resampling	pulse.ts resample()	Linear interpolation of timestamped luminance samples onto a uniform 20 Hz temporal grid (50 ms spacing).	Eliminates Android UI thread capture jitter, fulfilling the strict equidistant time requirement for digital IIR filtering.
4. Outlier Clamping	pulse.ts clampMotionSpikes()	Derivative thresholding: clamps step changes exceeding 3.5× the median absolute first difference.	Suppresses mechanical motion artifacts caused by transient finger slip before they induce filter ringing.
5. Bandpass Filter	pulse.ts applyBandpass()	Zero-mean centering + 2nd-order Butterworth cascaded highpass (0.7 Hz) and lowpass (3.5 Hz) IIR biquad filters.	Isolates cardiac pulsatile frequencies (42?210 BPM); rejects DC respiratory baseline wander and high-frequency thermal noise.
6. Optical Inversion	pulse.ts analyzeSignalWindow()	Signal polarity inversion: s[n] = -filtered[n]; discards first 0.8s filter settling transient.	Beer-Lambert Law alignment: systolic capillary blood expansion absorbs more light (luminance trough); inversion makes pulses positive peaks.
7. Peak & Beat Detection	pulse.ts analyzeSignalWindow()	Local maxima search with refractory distance constraint (? 0.35s / 7 samples at 20 Hz) and adaptive prominence (? + 0.25?).	Extracts true systolic peaks while preventing double-counting of dirotic notches or early diastolic reflections.
8. Interval Filtering	pulse.ts analyzeSignalWindow()	Derives RR intervals; rejects intervals with instantaneous BPM outside [40, 200] or deviating > 40% from median interval.	Guarantees high-fidelity Inter-Beat Interval (IBI) series for downstream autonomic HRV computation.
9. Biomarker Extraction	pulse.ts analyzeSignalWindow()	Computes AC/DC ratios of parallel red and green arrays, calculates HRV time-domain variance, and extracts morphological upstroke features.	Translates digital signal features into final clinical metrics including SpO2, sympathovagal balance (LF/HF), and pulse contour hemodynamics (Stroke Volume, Cardiac Output).

2. Mathematical Formulas & Algorithmic Formulations

Every formula utilized across the mobile application and research testbench is itemized below with exact parameter definitions.

2.1 Optical Tissue Verification & Subsampled Luminance

Code Location: mobile/src/pulse.ts (Lines 75–95)

```
avgRed = (1 / N) * sum(I_red[i * 4 * stride]),    avgGreen = (1 / N) * sum(I_green[i * 4 * stride + 1])
Valid Tissue = (avgRed >= 95) and (avgRed >= 1.15 * avgGreen)
```

Where stride = (width * height > 40,000) ? 4 : 1. Verifies the optical spectral ratio of oxygenated hemoglobin in dermal tissue versus stray ambient room lighting.

2.2 Uniform Temporal Resampling (Linear Interpolation)

Code Location: mobile/src/pulse.ts (Lines 609–624)

```
y(t_k) = y(t_i) + [ y(t_{i+1}) - y(t_i) ] * [ (t_k - t_i) / (t_{i+1} - t_i) ]
where t_k = t_0 + (k / f_s),    f_s = 20 Hz,    k in [0, floor(duration * f_s)]
```

2.3 Derivative-Based Outlier Clamping (Motion Artefact Reduction)

Code Location: mobile/src/pulse.ts (Lines 546–563)

```
delta_x[i] = |x[i] - x[i-1]|,    medDiff = median({delta_x[i]})
T_clamp = max(3.0, 3.5 * medDiff)
x*[i] = x*[i-1] + sign(x[i] - x*[i-1]) * T_clamp    [if |x[i] - x*[i-1]| > T_clamp]
```

2.4 2nd-Order Butterworth Bandpass IIR Filter (Biquad Formulation)

Code Location: mobile/src/pulse.ts (Lines 565–607) & research/deep_ppg_pipeline.py

```
Bilinear Transform Design (f_c in {0.7 Hz, 3.5 Hz}, f_s = 20 Hz, d = sqrt(2)):
theta = (pi * f_c) / f_s
beta = 0.5 * [ (1 - (d/2)*sin(theta)) / (1 + (d/2)*sin(theta)) ]
gamma = (0.5 + beta) * cos(theta)
Lowpass: alpha = (0.5 + beta - gamma)/2, b = [alpha, 2*alpha, alpha], a = [1, -2*gamma, 2*beta]
Highpass: alpha = (0.5 + beta + gamma)/2, b = [alpha, -2*alpha, alpha], a = [1, -2*gamma, 2*beta]
Difference Equation: y[n] = b_0*x[n] + b_1*x[n-1] + b_2*x[n-2] - a_1*y[n-1] - a_2*y[n-2]
```

2.5 Systolic Peak Detection with Refractory Beat Constraint

Code Location: mobile/src/pulse.ts (Lines 190–205)

```
Candidate Peak at index i satisfies:
s[i] > s[i-1] and s[i] > s[i+1] and s[i] > (mean_s + 0.25 * std_s)
Refractory Distance: (i - peak_last) >= floor(f_s * 0.35) = floor(20 * 0.35) = 7 samples (max ~171.4 BPM)
```

2.6 Heart Rate (BPM) and Dynamic Instantaneous Range

Code Location: mobile/src/pulse.ts (Lines 226–276, 357)

```
RR[k] = (peak[k] - peak[k-1]) / f_s    [seconds]
Heart Rate = round( 60.0 / median({RR_clean}) )
HR_min = round( min_k (60.0 / RR[k]) ),    HR_max = round( max_k (60.0 / RR[k]) )
```

2.7 Heart Rate Variability (HRV): RMSSD, SDNN, pNN50 & pNN20

Code Location: mobile/src/pulse.ts (Lines 242-265, 388-396)

```
RMSSD = sqrt( (1 / (M - 1)) * sum (RR[k+1] - RR[k])^2 ) * 1000 [ms]
SDNN = sqrt( (1 / M) * sum (RR[k] - mean(RR))^2 ) * 1000 [ms]
pNN50 = [ (Count of |RR[k+1] - RR[k]| > 50 ms) / (M - 1) ] * 100 [%]
pNN20 = [ (Count of |RR[k+1] - RR[k]| > 20 ms) / (M - 1) ] * 100 [%]
```

2.8 Autonomic Stress Score & Autonomic State Classification

Code Location: mobile/src/pulse.ts (Lines 278–284, 363)

```
Stress Score = clamp( 15 + max(0, 50 - RMSSD) * 1.8, 10, 95 ) [%]
State = 'Rest (Calm)' if RMSSD >= 42 ms; 'Elevated Stress' if RMSSD <= 22 ms; else 'Moderate'
```

2.9 Systolic Upstroke Time (Foot-to-Peak) & Vascular Elasticity

Code Location: mobile/src/pulse.ts (Lines 288–308, 349–355)

```
foot_idx = argmin_{j in [peak - floor(0.4*f_s), peak]} s[j]
T_up = ( (peak - foot_idx) / f_s ) * 1000 [ms]
Optimal Elasticity: T_up >= 130 ms | Moderate: 105-129 ms | Elevated Stiffness: T_up < 105 ms
```

2.10 Reflection Index (Digital Volume Pulse Morphology)

Code Location: mobile/src/pulse.ts (Lines 309–339)

```
Reflection Index (RI %) = [ (h_diastolic - h_foot) / (h_systolic - h_foot) ] * 100
Research Proxy: RI_proxy = clip( 1.0 - (T_up / 250.0), 0.20, 0.95 )
```

2.11 Perfusion Index (Pulsatile to Non-Pulsatile Ratio)

Code Location: mobile/src/pulse.ts (Lines 341–346)

```
Perfusion Index (PI %) = ( AC / DC ) * 100 = [ (max(s_pulsatile) - min(s_pulsatile)) / mean(x_unfiltered) ] * 100
```

2.12 Respiration Rate via Respiratory Sinus Arrhythmia (RSA)

Code Location: mobile/src/pulse.ts (Lines 451–486)

```
Breath Signal = ButterworthLowpass(x_pre_bandpass - mean(x), f_c = 0.5 Hz)
Respiration Rate = round( [ (N_breath_peaks - 1) / (t_last_peak - t_first_peak) ] * 60 ) [breaths/min]
Plausible Resting Range: 6 <= RR_resp <= 30 breaths/min
```

2.13 Calibrated Blood Pressure Estimation (One-Point Cuff Anchored)

Code Location: mobile/src/bpCalibration.ts (Lines 60–81)

```
delta_T_up = T_{up,cal} - T_{up,current} [ms],      delta_HR = HR_{current} - HR_{cal} [bpm]
delta_Sys = clamp( 0.50 * delta_T_up + 0.15 * delta_HR, -20, +20 )
delta_Dia = clamp( 0.25 * delta_T_up + 0.08 * delta_HR, -15, +15 )
Systolic = round( Sys_cal + delta_Sys ),      Diastolic = min( round( Dia_cal + delta_Dia ), Systolic - 15 )
Mean Arterial Pressure (MAP) = round( (2 * Diastolic + Systolic) / 3 )
Pulse Pressure (PP) = Systolic - Diastolic
Recalibration Rule: Flagged stale at >30 days; estimation refused at >60 days.
```

2.14 Longitudinal Biomarker Trends (Ordinary Least Squares)

Code Location: mobile/src/trendAnalysis.ts (Lines 34–41)

```
x[i] = (t[i] - t[0]) / (365.25 * 24 * 3600 * 1000) [years]
Slope (beta) = sum_{i=1}^N (x[i] - mean(x))*(y[i] - mean(y)) / sum_{i=1}^N (x[i] - mean(x))^2 [units/year]
```

2.15 Blood Oxygen Saturation (SpO2)

Code Location: mobile/src/pulse.ts (Lines 376-386)

```
Ratio (R) = (AC_red / DC_red) / (AC_green / DC_green)
SpO2 = round(110 - 25 * R) [%]
```

2.16 LF/HF Ratio (Frequency Domain HRV Surrogate)

Code Location: mobile/src/pulse.ts (Lines 398-406)

```
LF_var = max(0, SDNN^2 - RMSSD^2)
HF_var = RMSSD^2
LF/HF Ratio = LF_var / HF_var
```

2.17 Stroke Volume (SV), Cardiac Output (CO) & Cardiac Workload (RPP)

Code Location: mobile/src/pulse.ts (Lines 461-468)

```
Pulse Pressure (PP) = Systolic_BP - Diastolic_BP
Stroke Volume (SV) = round( PP * 1.5 ) [mL]
Cardiac Output (CO) = (SV * BPM) / 1000 [L/min]
Cardiac Workload (RPP) = BPM * Systolic_BP [mmHg·bpm]
```

3. Scientific Validation & Direct Research Paper Citations

The mathematical principles, optical filter thresholds, and physiological interpretations implemented in the codebase are grounded in and validated by peer-reviewed biomedical engineering literature:

Biomarker / Formula	Physiological Grounding	Primary Literature Citation	Scientific Validation Findings
Photoplethysmography & Beer-Lambert Law $s[n] = -y[n]$	Volumetric expansion of arteriolar vessels during ventricular systole increases light absorption by oxy- and deoxyhemoglobin, reducing backscattered luminance.	Allen, J. (2007). <i>Physiological Measurement</i> , 28(3), R1–R39. DOI: 10.1088/0967-3334/28/3/R01	Validates that optical finger transillumination reliably tracks cardiac pulse waves (AC pulsatile component riding on static DC tissue baseline).
Smartphone Camera PPG (cPPG) Flash + CMOS sensor capture	CMOS image sensors capture micro-fluctuations in reflected red luminance caused by pulsatile dermal perfusion.	Scully, C. G., et al. (2012). <i>IEEE TBME</i> , 59(2), 303–306. DOI: 10.1109/TBME.2011.2163157	Validated smartphone camera PPG against standard clinical ECG and finger pulse oximetry, obtaining correlation coefficients $r \geq 0.98$ for heart rate.
Butterworth Bandpass (0.7–3.5 Hz) Cascaded biquad IIR filter	Isolates cardiac fundamental and harmonic frequencies (42–210 BPM) while stripping respiratory drift (<0.5 Hz) and sensor noise (>4 Hz).	Elgendi, M. (2012). <i>Current Cardiology Reviews</i> , 8(1), 14–25. DOI: 10.2174/157340312801215782	Demonstrates that 2nd-order to 4th-order Butterworth bandpass filtering (0.5–3.5 Hz) produces maximal signal-to-noise ratio for peak-to-peak interval detection.
Time-Domain HRV (RMSSD, SDNN, pNN50) Successive difference formulas	RMSSD indexes high-frequency parasympathetic (vagal tone) cardiac control. SDNN indexes total autonomic regulatory capacity. pNN50 mirrors vagal modulation.	Task Force of ESC & NASPE (1996). <i>Circulation</i> , 93(5), 1043–1065. DOI: 10.1161/01.CIR.93.5.1043	Established international gold-standard clinical definitions and physiological significance for RMSSD, SDNN, and pNN50 in cardiovascular autonomic monitoring.
Systolic Upstroke Time (T_up) & Elasticity Foot-to-peak crest time	Moens-Korteweg Law ($c = \sqrt{Eh/2r\rho}$). Vessel stiffening increases pulse wave velocity (PWV), causing premature wave reflections that accelerate upstroke ($T_{up} < 105$ ms).	Chowienczyk, P. J., et al. (1999). <i>JACC</i> , 34(7), 2007–2014. DOI: 10.1016/S0735-1097(99)00444-1	Confirmed that digital pulse wave contour changes and accelerated systolic upstroke serve as reliable non-invasive surrogates for vascular endothelial dysfunction and arterial stiffness.
Reflection Index (RI %) & DVP Contour Diastolic/systolic peak ratio	Quantifies the secondary reflection wave originating from the lower body impedance mismatch at the iliac bifurcation returning to the upper limb.	Millasseau, S. C., et al. (2002). <i>Clinical Science</i> , 103(4), 371–377. DOI: 10.1042/cs1030371	Demonstrated that the ratio of diastolic peak height to systolic peak height (Reflection Index) directly correlates with systemic arterial vascular resistance.
Perfusion Index (PI %) AC / DC * 100%	Ratio of dynamic pulsatile blood volume (AC) to non-pulsatile background tissue/capillary bed volume (DC).	Lima, A., & Bakker, J. (2005). <i>Intensive Care Medicine</i> , 31(10), 1316–1326. DOI: 10.1007/s00134-005-2794-2	Validated Perfusion Index as a clinical marker for peripheral vascular tone, vasoconstriction, and circulatory integrity in medical telemetry.
Respiration Rate via RSA Lowpass baseline extraction	Respiratory Sinus Arrhythmia: Intrathoracic pressure changes during inspiration and expiration modulate venous return, cardiac stroke volume, and autonomic vascular tone.	Karlen, W., et al. (2013). <i>IEEE TBME</i> , 60(7), 1946–1953. DOI: 10.1109/TBME.2013.2246160	Demonstrated that filtering respiratory baseline modulation (0.1–0.5 Hz) from finger PPG accurately extracts respiration rate compared to capnometry.
Cuff-Anchored Calibrated BP Sensitivity perturbation model	Optical cPPG cannot derive absolute hydrostatic pressure (mmHg) without a reference anchor. Perturbation models track changes relative to known cuff readings.	Mukkamala, R., et al. (2015). <i>IEEE TBME</i> , 62(8), 1879–1901. IEEE Std 1708-2014	Mandates that cuffless PPG-based blood pressure estimates require periodic cuff recalibration to bound sensor drift and remain physiologically accurate.
Subject-Independent ML Validation GroupKFold on subject IDs	Vascular morphology acts as a unique biometric fingerprint. Random splits cause cross-validation data leakage where models memorize subjects rather than learning general pathology.	Saeb, S., et al. (2017). <i>GigaScience</i> , 6(5), gix019. DOI: 10.1093/gigascience/gix019	Proves that record-wise random splits in physiological data yield falsely inflated accuracies; mandates subject-level cross-validation (GroupKFold / LOSO).
Blood Oxygen Saturation (SpO2)	The ratio of pulsatile to baseline light absorption (AC/DC) across Red and Green wavelengths isolates oxygenated hemoglobin saturation.	Karlen, W., et al. (2010). <i>Conference of the IEEE Engineering in Medicine and Biology</i> . DOI: 10.1109/IEMBS.2010.5627622	Supports mobile-phone pulse oximeter approaches based on optical pulse signals and wavelength-dependent absorption.
LF/HF Ratio	Quantifies the sympathovagal balance, acting as a clinical surrogate for physical and mental stress states.	Malliani, A., et al. (1991). <i>Circulation</i> , 84(2), 482–492.	Frequency-domain analysis of HRV relates low- and high-frequency components to cardiovascular autonomic regulation.
Stroke Volume (SV) & Cardiac Output (CO)	Pulse contour analysis models (e.g., Liljestrand-Zander) establish that arterial pulse pressure is proportionally related to stroke volume.	Wang, L., et al. (2018). <i>Sensors</i> , 18(1), 173.	Describes noninvasive cardiac output estimation using a novel photoplethysmogram index and supports PPG-based cardiac output estimation approaches.

4. Regulatory Boundaries & Software as a Medical Device (SaMD) Notice

Clinical Boundary & Non-Diagnostic Scope: The optical photoplethysmography pipeline documented herein operates exclusively as a personal wellness and non-invasive physiological screening tool. It does not measure blood glucose or invasive arterial hemodynamics directly. In accordance with FDA / CE SaMD guidance:

- 1. Metabolic Surrogates:** Advanced Glycation End-products (AGEs) alter arterial wall stiffness and autonomic vagal tone over time, but PPG cannot replace chemical glucose testing. Patients must never adjust insulin or hypoglycemic medication based on optical estimations.
- 2. Blood Pressure Grounding:** Blood pressure estimates are one-point cuff-anchored perturbation models, not direct hydrostatic measurements, and require regular recalibration.
- 3. Diagnostic Authority:** Users experiencing symptoms or seeking clinical diagnoses must consult a licensed physician and undergo standardized laboratory blood tests.